VIT BHOPAL

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PROJECT REPORT

Title:- Student Performance Prediction

INTRODUCTION:-

In education it is really helpful to know how well a student will do in school. This helps us find students who are struggling and give them the help they need to do better. This project uses Machine Learning, which's like a computer program that learns and something called Linear Regression to guess what marks a student will get at the end of the year. It looks at how a student studies and how they did on tests before to make this guess, about the students marks.

Objectives:-

To build a model that predicts final grade (G3) of students

To analyze the effect of:

- Study time
- Absences
- Previous marks (M1, M2)

To classify students as Pass/Fail and assign grades

Dataset Description:-

- Dataset used: student-mat.csv
- Source: UCI Machine Learning Repository
- Key features used:

Feature	Description
Studytime	Weekly study time (1–8 scale)
Absences	Number of absences
G1 (M1)	First period marks
G2 (M2)	Second period marks
G3	Final marks (target variable)

Technologies Used:-

- Python
- Libraries:
- pandas → Data handling
- sklearn → Machine learning
- train_test_split
- LinearRegression

Methodology:-

Step 1: Data Loading

- Dataset is loaded using pandas

- Semicolon (;) is used as separator

Step 2: Data Preprocessing

- Selected relevant columns:studytime, absences, G1, G2, G3
- Renamed:G1 → M1

  G2 → M2

Step 3: Feature Selection

- Input features (X):studytime, absences, M1, M2
- Target variable (y):G3

Step 4: Data Splitting

- 80% → Training data
- 20% → Testing data

Step 5: Model Training

- Used Linear Regression model
- Model learns relationship between inputs and final marks

Step 6: Prediction

- User inputs:Study time,Absences,M1, M2 marks
- Model predicts final marks (G3)

Working of the Model:-

Linear Regression follows the equation:

$$G3=b0+b1(studytime)+b2(absences)+b3(M1)+b4(M2)$$

- The model calculates coefficients (b1, b2, etc.)
- Uses them to predict final marks

Output System:-

✅ Pass/Fail Criteria:

Pass: Marks ≥ 33

Fail: Marks < 33

Sample Output:-

Enter student details:

Study time: 5

Absences: 2

M1: 70

M2: 75

Predicted Final Marks: 74.5

Result: PASS

Grade: B (Very Good)

Advantages:-

- Simple and easy to implement
- Helps teachers identify weak students
- Uses real data for prediction
- Fast prediction

Limitations:-

- Linear relationship is assumed by linear regression
- Data set quality decides quality
- We used Limited features
- Does not consider external factors (health, environment)

Future Scope:-

- I used advanced models like Random Forest, Neural Networks
- I Add more features like internet usage, family background
- I will Deploy as a web app or mobile app
- Improve accuracy with larger datasets

Challenges Faced:-

- Understanding dataset format (semicolon-separated CSV and selecting correct columns)
- Data preprocessing issues, such as renaming columns and choosing relevant features
- Model accuracy limitations due to simple Linear Regression
- Handling user input properly and ensuring correct data types
- Overfitting/underfitting concerns due to limited features
- Interpreting model output and converting it into meaningful results (grades, pass/fail)

Conclusion:-

The project shows that Machine Learning is really good at predicting how well students will do. The Machine Learning Linear Regression model is a way to figure out what marks students will get at the end and which group they will be

in. This Machine Learning system helps teachers do something, about it so students can do better.

What I Learned from This Project:-

- Basics of Machine Learning workflow
- Practical use of Linear Regression algorithm
- Importance of feature selection in improving predictions
- How to use Python libraries (pandas, sklearn)
- Understanding how real-world data affects model performance
- Converting predictions into useful decision-making outputs (grades, result system)

References:-

- UCI Machine Learning Repository
- Scikit-learn Documentation
- Python Pandas Documentation